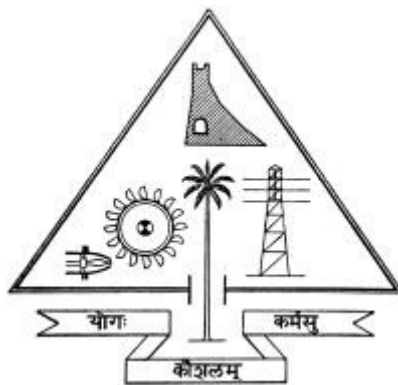


COUNTERPASS: A SPATIO-TEMPORAL COUNTERFACTUAL FRAMEWORK FOR EVALUATING ALTERNATIVE PASSING DECISIONS IN FOOTBALL VIDEO

*Mini Project Report submitted in partial fulfillment of the requirements for the
award of the degree of **Master of Computer Applications** of the **APJ Abdul
Kalam Technological University***

Submitted by

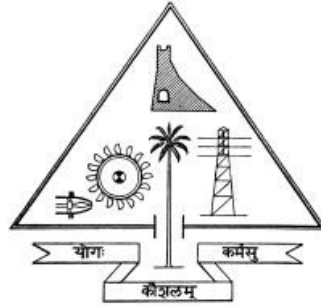
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CERTIFICATE

*This is to certify that the mini project titled **“COUNTERPASS: A SPATIO-TEMPORAL COUNTERFACTUAL FRAMEWORK FOR EVALUATING ALTERNATIVE PASSING DECISIONS IN FOOTBALL VIDEO”** is a bonafide work done by **AKARSH JAISON (TCR25MCA-2009)** under my supervision and guidance, and is submitted in November 2026 in partial fulfillment of the requirements for the award of the Degree of Master of Computer Applications from APJ Abdul Kalam Technological University (KTU).*

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ABSTRACT

This project presents **CounterPass: A Spatio-Temporal Counterfactual Framework for Evaluating Alternative Passing Decisions in Football Video**, an AI-powered football analytics system that combines computer vision, multi-object tracking, graph-based spatial modelling, and counterfactual reasoning to analyse passing decisions from broadcast football videos. The proposed framework automatically detects players and the ball using the YOLO object detection model, tracks their movements throughout the match using ByteTrack, and reconstructs the spatio-temporal relationships between players as a dynamic interaction graph. CounterPass generates multiple counterfactual passing scenarios by identifying feasible alternative receivers available at the instant of each pass. Every candidate pass is evaluated using spatial metrics such as passing lane openness, defensive pressure, teammate availability, distance, interception probability, and progression towards the attacking goal. The framework computes a composite tactical score for every alternative and compares it with the actual pass played, thereby providing objective insights into whether a better passing decision existed.

The generated analysis is presented through annotated match videos, interactive tactical visualisations, and comprehensive analytical reports that explain the strengths and weaknesses of each passing decision. The proposed framework offers coaches, analysts, and players an affordable and intelligent football performance analysis solution without requiring expensive GPS trackers, wearable sensors, or multi-camera installations. By integrating Artificial Intelligence, Computer Vision, and spatio-temporal reasoning, CounterPass aims to improve tactical awareness, support post-match analysis, and promote data-driven decision making in modern football.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CV	Computer Vision
CUDA	Compute Unified Device Architecture
DL	Deep Learning
FPS	Frames Per Second
GPS	Global Positioning System
GUI	Graphical User Interface
HTML	HyperText Markup Language
IDE	Integrated Development Environment
IoU	Intersection over Union
JSON	JavaScript Object Notation
ML	Machine Learning
MOT	Multi Object Tracking
OCR	Optical Character Recognition
REST	Representational State Transfer
SQL	Structured Query Language
UI	User Interface
YOLO	You Only Look Once
GPU	Graphics Processing Unit
CSV	Comma Separated Values
GNN	Graph Neural Network
STG	Spatio-Temporal Graph

CHAPTER 1

INTRODUCTION

The rapid advancement of Artificial Intelligence (AI) and Computer Vision (CV) has transformed the field of sports analytics. Modern deep learning models are capable of automatically detecting players, tracking movement trajectories, recognizing game events, and extracting tactical information directly from football videos. Professional clubs increasingly employ data-driven analytical systems to evaluate player performance, study opposition tactics, and support coaching decisions. However, most existing football analytics platforms primarily provide descriptive statistics such as possession percentage, player heat maps, passing networks, and movement trajectories. While these visualizations assist coaches in understanding match performance, they seldom evaluate whether a player could have made a better passing decision at a particular moment.

One of the major limitations of existing football video analysis systems is the absence of counterfactual tactical reasoning. Traditional analytics explain what happened during a match but rarely investigate what could have happened if a different decision had been taken. Counterfactual analysis enables the evaluation of alternative scenarios by comparing the actual action performed with several feasible alternatives under identical match conditions. In football, such analysis can reveal whether another teammate was in a better position to receive the ball, whether a different passing lane would have reduced interception risk, or whether an alternative pass could have produced a greater attacking advantage.

Recent developments in object detection and multi-object tracking have made automated tactical analysis increasingly feasible. State-of-the-art object

detection models such as YOLO accurately identify players and the football in broadcast videos, while tracking algorithms such as ByteTrack maintain consistent player identities throughout the match. These technologies provide reliable spatial and temporal information describing player positions and movements over time. However, positional information alone is insufficient for intelligent tactical evaluation. Effective passing analysis requires a deeper understanding of player interactions, defensive pressure, spatial occupation, passing lane availability, and future game progression.

To address these limitations, this project proposes **CounterPass: A Spatio-Temporal Counterfactual Framework for Evaluating Alternative Passing Decisions in Football Video**. The proposed system integrates Artificial Intelligence, Computer Vision, graph-based spatial modelling, and counterfactual reasoning into a unified football analytics framework. The system processes ordinary football match videos, automatically detects players and the ball using the YOLO object detection model, tracks player movements using ByteTrack, and constructs a dynamic spatio-temporal representation of the match that continuously models player interactions.

Unlike conventional football analysis tools that simply visualize completed passes, CounterPass evaluates every passing decision by generating multiple feasible alternative passing scenarios. For each pass, the framework identifies all potential receiving teammates and evaluates every candidate using several tactical factors, including passing lane openness, defensive pressure, teammate positioning, distance to goal, interception probability, player movement direction, and spatial progression towards attacking areas. Each alternative is assigned a tactical score, allowing direct comparison between the actual pass played and the most advantageous alternative available at that instant.

The framework further enhances tactical understanding by employing spatio-temporal graph modelling to represent player relationships throughout the game. Each player is represented as a node within a dynamic graph, while edges represent potential passing opportunities influenced by continuously

changing tactical conditions. This graph representation enables the system to analyse the evolving structure of the game rather than evaluating isolated player positions. Counterfactual reasoning is then applied to estimate how alternative passing decisions might have influenced subsequent attacking opportunities, thereby providing coaches and analysts with actionable tactical insights instead of descriptive statistics alone.

The results generated by CounterPass are presented through annotated football videos, tactical visualisations, passing opportunity overlays, player interaction graphs, and comprehensive analytical reports. These outputs allow coaches, analysts, and players to understand why a particular passing option was selected, how alternative passes compare, and what tactical advantages or disadvantages each decision produces. Such insights can significantly improve player awareness, coaching strategies, post-match reviews, and tactical training sessions.

1.1 Objectives

The primary objective of CounterPass is to develop an intelligent football analytics framework capable of evaluating passing decisions directly from football match videos using Artificial Intelligence and Computer Vision techniques.

The specific objectives of the proposed framework are as follows:

- To automatically detect football players, referees, and the ball from ordinary football match videos using the YOLOv11 object detection model.
- To track player movements throughout the duration of the match using multi-object tracking algorithms such as ByteTrack.
- To construct a dynamic spatio-temporal representation of player interactions during football matches.
- To generate counterfactual passing scenarios by identifying alternative teammates who could have received the ball.

- To evaluate every candidate pass using tactical parameters including passing lane openness, defensive pressure, interception probability, and progression towards goal.
- To provide explainable tactical recommendations that support coaching decisions and player development.
- To generate annotated videos, visualizations, and analytical reports that facilitate post-match analysis.
- To provide an affordable football analytics solution that does not require wearable sensors, GPS trackers, or specialized camera infrastructure.

CHAPTER 2

LITERATURE REVIEW

Football, being one of the most widely played sports globally, has witnessed substantial advancements in video-based analysis techniques for player tracking, event detection, and tactical evaluation. Modern football clubs increasingly depend on data-driven systems to improve player performance, evaluate team strategies, and support coaching decisions. However, despite considerable progress in football analytics, intelligent tactical decision support remains an active area of research.

One of the fundamental research directions in football analytics involves automatic player and ball detection from video sequences. Earlier approaches primarily relied on image processing techniques such as background subtraction, color segmentation, and handcrafted feature extraction. Although these methods demonstrated reasonable performance under controlled environments, they struggled with varying lighting conditions, player occlusions, and camera movements. The introduction of deep learning-based object detection techniques has substantially improved detection accuracy and enabled real-time sports analytics applications. The YOLO (You Only Look Once) family of object detectors, originally proposed by Redmon et al., demonstrated that object localization and classification could be performed simultaneously with remarkable efficiency [1]. Recent improvements in the YOLO architecture, particularly YOLOv11, have further enhanced small object detection and inference speed, making the model highly suitable for football video analysis [2].

Multi-object tracking has emerged as another critical component of football analytics systems. While object detection identifies players in

individual frames, tracking algorithms preserve player identities across consecutive frames to reconstruct movement trajectories. Recent studies have demonstrated the effectiveness of modern tracking algorithms such as DeepSORT and ByteTrack in dynamic environments involving multiple moving objects. ByteTrack, proposed by Zhang et al., significantly improves tracking robustness by associating both high-confidence and low-confidence detections, thereby reducing identity switching and maintaining trajectory continuity throughout football matches [3]. Accurate player tracking forms the basis for advanced analytical tasks including formation analysis, passing networks, and tactical evaluation.

Several researchers have explored video-based football analysis systems capable of extracting tactical information directly from football videos. A video-based tracking system for football player analysis using efficient convolution operators demonstrated the feasibility of combining convolutional techniques with player localization to generate movement trajectories suitable for post-match analysis. Similarly, recent evaluations of football player and ball tracking systems have highlighted the importance of balancing detection accuracy with computational efficiency in real-world environments [6]. Although these systems provide valuable positional information, they primarily focus on player tracking and do not investigate alternative tactical decisions or passing opportunities.

Graph-based approaches have gained considerable attention in recent years for modelling interactions among football players. In graph-based representations, players are treated as nodes while passing relationships and spatial interactions are represented as edges. Such representations preserve both spatial and temporal information and provide a richer understanding of tactical behaviour compared to traditional statistical approaches. Graph Attention Networks (GAT), introduced by Veličković et al., have demonstrated significant potential in learning complex relationships within dynamic graph structures [8]. These techniques have subsequently influenced several sports analytics applications involving tactical modelling, movement prediction,

and player interaction analysis.

Another emerging research direction is counterfactual reasoning in Artificial Intelligence. Counterfactual analysis focuses on answering “what-if” questions by evaluating how alternative decisions could have influenced observed outcomes. Unlike conventional predictive models, counterfactual systems provide explainable insights by comparing actual events with plausible alternatives. Counterfactual explanations have become increasingly important in Explainable Artificial Intelligence (XAI) because they improve transparency and support informed decision-making [9]. Although counterfactual reasoning has been applied successfully in domains such as healthcare, finance, and autonomous systems, its application to football tactical analysis remains relatively unexplored.

Recent systematic reviews have demonstrated the growing role of Artificial Intelligence in sports performance analysis. AI-based techniques have been successfully applied for player tracking, injury prediction, tactical evaluation, performance forecasting, and match outcome prediction. Clemente’s systematic review highlights that sports analytics is rapidly transitioning from descriptive statistics towards intelligent decision-support systems capable of generating actionable recommendations [10]. These findings indicate a significant opportunity for integrating explainable AI techniques into football analytics to improve tactical understanding and player development.

In summary, the literature indicates substantial progress in football analytics, particularly in the areas of player detection, tracking, and performance evaluation. However, there remains a significant research gap in affordable, explainable, and AI-driven tactical analysis systems capable of evaluating alternative passing decisions directly from football videos. CounterPass addresses this gap by introducing a spatio-temporal counterfactual framework that supports data-driven coaching, tactical education, and post-match analysis.

CHAPTER 3

ENVIRONMENTAL STUDY

This chapter describes the hardware and software environment required for the successful development, deployment, and execution of the CounterPass framework. Since the proposed system performs computer vision, deep learning inference, multi-object tracking, and spatio-temporal tactical analysis, an appropriate computational environment is essential to ensure accurate and efficient performance. The selected technologies provide an optimal balance between computational efficiency, scalability, portability, and ease of development.

3.1 System Configuration

The CounterPass framework has been designed to operate on standard computing platforms while providing sufficient computational capability for football video analysis. The system processes football videos frame by frame, performs player and ball detection, tracks player movement, constructs spatio-temporal representations, evaluates counterfactual passing opportunities, and generates tactical reports.

3.1.1 Hardware Requirements

The minimum hardware configuration required to execute the proposed system is as follows:

- Processor: Intel Core i5 (10th Generation or higher) or AMD Ryzen 5 processor.

- Memory: Minimum 8 GB RAM (16 GB recommended for improved processing performance).
- Storage: At least 50 GB of available storage for football videos, trained models, and generated reports.
- Graphics Processing Unit: NVIDIA GPU supporting CUDA acceleration is recommended for faster deep learning inference, although CPU execution is also supported.
- Display: Full HD (1920 × 1080) monitor or higher resolution.

The recommended hardware configuration enables efficient execution of object detection models, multi-object tracking algorithms, and tactical analysis modules while maintaining acceptable processing speed for football videos.

3.1.2 Software Requirements

The software environment consists of the operating system, programming language, development tools, Artificial Intelligence frameworks, and supporting libraries required for implementing the CounterPass framework.

The software requirements are listed below:

- Operating System: Windows 11, Ubuntu 22.04 LTS, or macOS.
- Programming Language: Python 3.11 or later.
- Database: PostgreSQL or SQLite.
- Web Browser: Google Chrome, Mozilla Firefox, or Microsoft Edge.
- Version Control System: Git.
- Development Environment: Visual Studio Code.

3.1.3 Software Requirements for Development

The development environment includes several software frameworks and open-source libraries responsible for computer vision, tactical analysis, and system deployment.

The development software includes:

- Visual Studio Code as the Integrated Development Environment.
- Python virtual environment for dependency management.
- GitHub for source code version control.
- PyTorch deep learning framework.
- OpenCV image and video processing library.
- Ultralytics YOLO implementation.
- ByteTrack multi-object tracking framework.
- Flask backend framework.
- PostgreSQL database server.

These technologies provide an integrated development platform suitable for implementing modern AI-based football analytics applications.

3.2 Software Specification

The CounterPass framework combines Artificial Intelligence, Computer Vision, and graph-based tactical analysis into a unified system. Each software component performs a specific role within the overall processing pipeline.

3.2.1 Python

Python serves as the primary implementation language of the CounterPass framework. It coordinates every stage of the analysis pipeline, including

video processing, player detection, tactical evaluation, report generation, and communication between software modules.

Python provides extensive support for Artificial Intelligence through numerous open-source libraries while offering high developer productivity and portability across different operating systems.

3.2.2 YOLOv11

YOLOv11 functions as the primary object detection engine within the framework. It detects football players, referees, and the ball directly from each video frame.

Compared to earlier object detection techniques, YOLO performs localization and classification simultaneously, enabling high detection accuracy while maintaining real-time processing capability. Its efficiency makes it particularly suitable for football video analysis where multiple moving objects must be detected continuously throughout an entire match.

3.2.3 ByteTrack

ByteTrack is responsible for maintaining player identities throughout the video sequence after object detection has been completed.

Unlike conventional tracking algorithms that discard low-confidence detections, ByteTrack associates both high-confidence and low-confidence detections to preserve tracking continuity during player occlusions, rapid movement, and camera motion. This enables accurate reconstruction of player trajectories required for tactical analysis.

3.2.4 OpenCV

OpenCV provides image processing and video manipulation capabilities for the proposed system.

The library is responsible for reading football videos, extracting video frames, preprocessing images, rendering annotations, drawing tactical visualizations, highlighting passing lanes, and exporting the analysed output

videos.

Its extensive collection of computer vision algorithms makes OpenCV one of the fundamental components of the CounterPass framework.

3.2.5 *PyTorch*

PyTorch provides the deep learning infrastructure required to execute neural network models used within the framework.

It enables efficient model loading, GPU acceleration, tensor computation, and inference execution. PyTorch also allows future extension of the framework through custom model training and fine-tuning using football datasets.

3.2.6 *Flask*

Flask functions as the backend framework responsible for communication between the graphical user interface and the Artificial Intelligence modules.

It handles video uploads, initiates tactical analysis, manages analysis sessions, stores generated reports, and returns processed results to the frontend application.

Flask's lightweight architecture makes it suitable for integrating deep learning models into web-based applications.

3.2.7 *PostgreSQL*

PostgreSQL serves as the primary database management system of the CounterPass framework.

The database stores uploaded football videos, user information, generated tactical reports, system configurations, and historical analysis records.

Its reliability and scalability make it appropriate for future deployment in larger football analytics environments.

3.2.8 Visual Studio Code

Visual Studio Code is used as the Integrated Development Environment throughout the project.

It provides syntax highlighting, debugging tools, Git integration, Python support, extension management, and efficient project organization. Its extensive plugin ecosystem significantly simplifies development and testing.

3.2.9 Git

Git provides distributed version control for the entire software development process.

It maintains complete project history, supports collaborative development, enables rollback to previous versions, and ensures reliable source code management throughout implementation.

The use of Git also facilitates future expansion and maintenance of the CounterPass framework.

CHAPTER 4

SYSTEM ANALYSIS

In the field of football analytics, coaches and analysts continuously strive to improve team performance by studying player movements, tactical formations, and passing decisions. Traditionally, this analysis has been performed manually by reviewing recorded match videos. Although manual analysis provides valuable insights, it is time-consuming, subjective, and often fails to identify subtle tactical opportunities that occur within fractions of a second. Modern Artificial Intelligence and Computer Vision techniques have significantly improved player detection and tracking; however, existing systems primarily focus on describing what occurred during the match rather than evaluating whether alternative tactical decisions could have produced better outcomes.

The proposed system, CounterPass, addresses this limitation by introducing a spatio-temporal counterfactual framework capable of analysing passing decisions directly from football match videos. The framework combines player detection, multi-object tracking, graph-based tactical modelling, and counterfactual reasoning to evaluate not only the pass that was played but also multiple feasible alternative passing options. This enables coaches, players, and analysts to understand whether a better tactical decision existed at a particular moment and how that decision might have influenced the subsequent play.

4.1 Requirements Analysis

The primary requirement of the proposed framework is the ability to process football videos captured using ordinary broadcast cameras. Unlike many

commercial analytics platforms that require specialized tracking hardware or wearable GPS devices, CounterPass should operate directly on uploaded video files without any additional infrastructure.

Another important requirement is accurate player and ball detection. Since all subsequent analytical modules depend on object detection, the framework must reliably identify players, referees, and the football under varying lighting conditions, camera angles, player occlusions, and match situations. High detection accuracy is essential for maintaining reliable player trajectories throughout the game.

The system should also provide continuous multi-object tracking. Every detected player must retain a consistent identity across consecutive video frames so that meaningful movement trajectories can be reconstructed. These trajectories serve as the foundation for analysing player positioning, tactical formations, and passing opportunities.

A unique requirement of CounterPass is counterfactual tactical analysis. Instead of analysing only completed passes, the framework should identify multiple feasible receivers available at every passing event and evaluate each alternative using tactical metrics such as passing lane openness, defensive pressure, spatial occupation, interception probability, receiver positioning, and progression towards the opponent's goal. This capability distinguishes CounterPass from traditional football analytics systems.

Finally, the system should be modular and extensible. Future analytical modules such as formation recognition, offside detection, expected threat estimation, pressing analysis, and player performance evaluation should be integrated without requiring major architectural modifications.

4.2 Existing System

Most existing systems perform player detection using deep learning-based object detection algorithms and subsequently generate descriptive statistics such as possession percentage, pass completion rate, player heat maps, sprint distance, and movement trajectories. While these analytical tools provide

valuable post-match insights, they primarily focus on summarising historical events rather than evaluating tactical decision-making.

Professional football organisations commonly employ advanced analytics platforms such as Hudl, StatsBomb, Wyscout, and Catapult. These systems provide detailed match statistics and visualisations that assist coaches in understanding team performance. However, they generally rely on proprietary datasets, wearable tracking devices, manually annotated event data, or multiple synchronized camera systems. Such requirements considerably increase implementation costs and restrict accessibility for amateur football clubs, educational institutions, and independent researchers.

4.3 *Limitations of Existing System*

- Existing systems depend on expensive hardware such as GPS trackers, wearable sensors, and multi-camera setups.
- Most football analytics platforms focus primarily on descriptive statistics rather than tactical decision support.
- Many systems rely heavily on manually annotated datasets, increasing time and operational costs.
- Existing solutions provide limited support for evaluating alternative passing opportunities.
- Commercial football analytics platforms are often inaccessible to amateur clubs and educational institutions due to high subscription costs.
- Most systems lack explainable Artificial Intelligence capabilities and fail to justify their recommendations.
- Current approaches generally analyze only completed events and do not perform counterfactual reasoning.

- Player tracking accuracy may be affected by occlusions, rapid movement, and varying lighting conditions.
- Existing football analytics tools require proprietary infrastructure that is not widely available.
- Very few systems integrate object detection, tracking, graph modelling, and tactical evaluation within a single framework.

4.4 Proposed System

To overcome the limitations of existing football analytics systems, this project proposes **CounterPass: A Spatio-Temporal Counterfactual Framework for Evaluating Alternative Passing Decisions in Football Video**. The proposed framework integrates Artificial Intelligence, Computer Vision, Multi-Object Tracking, Graph-based Tactical Modelling, and Counterfactual Reasoning into a unified football analysis platform.

The system begins by allowing users to upload ordinary football match videos through a simple graphical interface. Each uploaded video is processed frame by frame using the YOLOv11 object detection model, which identifies players, referees, and the football with high accuracy. The detected objects are then passed to the ByteTrack tracking algorithm, which maintains unique identities for every player throughout the match.

Once player trajectories have been reconstructed, the framework constructs a dynamic spatio-temporal graph representing player interactions. Every player is represented as a graph node, while potential passing relationships form weighted graph edges. Since player positions continuously change throughout the match, the graph is updated dynamically for every video frame, enabling the framework to capture the evolving tactical structure of the game.

Whenever a passing event occurs, CounterPass generates multiple counterfactual scenarios by identifying every teammate who could realistically receive the ball. Each candidate pass is evaluated using a

combination of tactical metrics, including passing lane openness, defensive pressure, receiver positioning, available attacking space, passing distance, interception probability, player movement direction, and progression towards the opponent's goal.

Based on these factors, the system calculates a tactical score for every alternative passing option. The actual pass executed during the match is then compared with the highest-ranked alternatives, enabling coaches and analysts to determine whether a more effective passing decision existed.

The final analysis is presented through annotated football videos, tactical visualisations, passing lane overlays, player trajectory maps, and detailed analytical reports. These visual outputs allow users to understand not only what happened during the match but also why an alternative tactical decision may have produced a better outcome.

4.5 Advantages of Proposed System

- Provides automatic football video analysis using Artificial Intelligence and Computer Vision.
- Detects players, referees, and the ball accurately using the YOLOv11 object detection model.
- Maintains continuous player identities using ByteTrack for reliable movement tracking.
- Generates counterfactual passing scenarios to evaluate alternative tactical decisions.
- Utilizes spatio-temporal graph modelling for analysing player interactions and passing opportunities.
- Produces explainable tactical recommendations supported by visual evidence.
- Generates annotated videos, tactical visualizations, and comprehensive analytical reports.

- Eliminates the need for wearable sensors, GPS devices, and specialized camera infrastructure.
- Offers a cost-effective solution suitable for football academies, educational institutions, and amateur clubs.
- Supports future expansion through a modular and scalable software architecture.
- Enhances coaching decisions, player development, and post-match analysis.
- Promotes data-driven tactical evaluation in modern football analytics.

4.6 Functional Requirements

Functional requirements describe the services and operations that the CounterPass framework must perform in order to achieve its objectives. These requirements define the behaviour of every major module within the system and specify how users interact with the application. The proposed framework consists of several interconnected modules that work together to analyse football videos and generate intelligent tactical recommendations.

4.6.1 Video Upload Module

The Video Upload module serves as the entry point of the CounterPass framework. It enables users to upload football match videos captured from broadcast recordings, training sessions, or practice matches. The system accepts commonly used video formats such as MP4, AVI, and MOV while validating file integrity before processing. Once the upload is completed, the video is stored temporarily for analysis. The module also verifies video resolution and frame quality to ensure compatibility with the computer vision pipeline. Appropriate error messages are displayed whenever unsupported or corrupted files are detected.

4.6.2 Video Pre-processing Module

Before analysis begins, the uploaded video undergoes a preprocessing stage to improve the quality of subsequent computer vision tasks. During this stage, the video is divided into individual frames at a predefined frame rate. Noise reduction, frame resizing, colour normalization, and image enhancement techniques are applied wherever necessary. This preprocessing stage improves the robustness of player detection under varying lighting conditions and ensures consistent input for the deep learning models.

4.6.3 Player and Ball Detection Module

The Player and Ball Detection module is responsible for automatically identifying football players, referees, and the ball in every frame of the uploaded video. The proposed system employs the YOLOv11 object detection model because of its high detection accuracy and real-time inference capability. The module should accurately detect multiple moving objects simultaneously even under challenging situations such as player overlaps, rapid movement, partial occlusions, and camera motion. Reliable object detection forms the foundation for every subsequent analytical stage within the framework.

4.6.4 Player Tracking Module

Following player detection, the framework performs multi-object tracking using the ByteTrack algorithm. The primary objective of this module is to assign a unique identity to every detected player and preserve that identity throughout the entire video sequence. Continuous player tracking enables the system to reconstruct complete movement trajectories for every player, even during temporary occlusions or close interactions with other players. Accurate tracking is essential for analysing tactical formations, player positioning, and passing opportunities.

4.6.5 Spatio-Temporal Graph Construction Module

The Spatio-Temporal Graph Construction module models the football match as a continuously evolving interaction network. Every player is represented as a graph node, while possible passing relationships between teammates are represented as weighted edges. The graph changes dynamically as players move across the field. This representation enables the framework to capture both the spatial arrangement of players and the temporal evolution of the match, providing significantly richer tactical information than isolated frame-by-frame analysis.

4.6.6 Counterfactual Pass Generation Module

One of the distinguishing features of CounterPass is its ability to generate counterfactual passing scenarios. Whenever a pass is detected, the framework identifies every teammate who could realistically have received the ball under the same match conditions. Rather than evaluating only the actual pass that occurred, the module generates multiple alternative passing possibilities that satisfy spatial and tactical constraints. These counterfactual scenarios enable objective comparison between the chosen decision and other feasible passing options.

4.6.7 Tactical Evaluation Module

The Tactical Evaluation module assesses every candidate passing option using multiple tactical metrics. These include passing lane openness, receiver availability, defensive pressure, passing distance, interception probability, movement direction, available attacking space, and progression towards the opponent's goal. Each factor contributes to a composite tactical score that reflects the quality of the passing opportunity. Based on these scores, the framework ranks all candidate passes and identifies the most effective tactical decision available at that instant.

4.6.8 Visualization Module

The Visualization module presents the analytical results in an intuitive graphical format. Player trajectories, recommended passing lanes, detected players, ball movement, and tactical annotations are superimposed directly on the original football video. Different colours and graphical indicators are used to distinguish the actual pass from the recommended alternatives. These visualizations help coaches and players understand complex tactical situations quickly without requiring extensive statistical interpretation.

4.6.9 Report Generation Module

After completing the tactical analysis, the framework automatically generates a detailed analytical report. The report summarizes player movements, passing decisions, tactical observations, recommended alternatives, and overall match insights. It also includes visual illustrations and statistical summaries that support coaching decisions and post-match analysis. The generated reports may be stored for future reference and performance comparison across multiple matches.

4.6.10 User Interface Module

The User Interface provides a simple and user-friendly environment through which coaches, analysts, researchers, and players can interact with the system. Users can upload videos, initiate analysis, monitor processing progress, visualize tactical recommendations, and download generated reports without requiring technical knowledge. The interface has been designed to minimize user effort while providing maximum accessibility.

4.6.11 Error Handling Module

The Error Handling module improves the reliability of the CounterPass framework by detecting and managing unexpected situations during execution. Invalid video formats, corrupted files, missing frames, insufficient system resources, and AI processing failures are handled gracefully by

displaying meaningful error messages to the user. Proper error handling prevents abrupt system termination and ensures a stable user experience during football video analysis.

4.7 Non-Functional Requirements

Non-functional requirements define the quality attributes of the CounterPass framework. Unlike functional requirements, which describe what the system should do, non-functional requirements specify how well the system should perform. These requirements ensure that the proposed system remains efficient, reliable, secure, and user-friendly throughout its operation.

4.7.1 Performance

The CounterPass framework should process football match videos efficiently while maintaining high analytical accuracy. The system should minimize the time required for video preprocessing, player detection, tracking, and tactical evaluation. GPU acceleration should be utilized whenever available to reduce processing time. Even when executed on standard computing hardware, the framework should complete analysis within a reasonable time without compromising the quality of generated results.

4.7.2 Reliability

The framework should consistently produce reliable tactical analysis for different football matches captured under varying environmental conditions. Minor variations in lighting, camera movement, player density, or video quality should not significantly affect the accuracy of player detection and tracking. Reliable performance ensures that coaches and analysts can confidently use the generated recommendations during tactical evaluation.

4.7.3 Accuracy

Accurate player detection and tracking form the foundation of the CounterPass framework. The object detection model should correctly identify

players, referees, and the football with minimal false detections. Similarly, the tracking algorithm should preserve player identities throughout the match while minimizing identity switches. The tactical evaluation module should generate recommendations that closely reflect realistic football scenarios and tactical principles.

4.7.4 Scalability

The proposed architecture should support future expansion without requiring major redesign. Additional analytical modules such as offside detection, expected threat analysis, formation recognition, pressing analysis, player fatigue estimation, and match prediction should be integrated seamlessly into the existing framework. The modular architecture ensures that future research developments can be incorporated efficiently.

4.7.5 Maintainability

The software should follow a modular design that simplifies maintenance and future development. Individual components such as player detection, tracking, tactical evaluation, and visualization should remain independent so that modifications to one module do not significantly affect others. Proper documentation and standardized coding practices further improve maintainability.

4.8 Feasibility Study

A feasibility study evaluates whether the proposed CounterPass framework can be successfully developed using available resources. It examines technical capability, operational practicality, economic viability, and project scheduling before implementation begins.

4.8.1 Technical Feasibility

The proposed system is technically feasible because it is built entirely upon mature and widely adopted open-source technologies. Python serves as the

primary programming language, while YOLOv11 provides accurate player detection and ByteTrack performs robust multi-object tracking. OpenCV supports video processing, PyTorch enables deep learning inference, and Flask facilitates backend integration.

The required software frameworks are actively maintained and supported by extensive developer communities. Furthermore, the computational requirements can be satisfied using modern personal computers equipped with multi-core processors and optional GPU acceleration. Therefore, the implementation of CounterPass is technically achievable within the scope of an MCA mini project.

4.8.2 Operational Feasibility

From an operational perspective, CounterPass offers significant practical advantages. The framework processes ordinary football match videos without requiring specialized cameras, wearable sensors, or proprietary tracking infrastructure. Coaches, analysts, students, and researchers can operate the system with minimal training through a simple graphical interface.

The generated tactical reports and visualizations are easy to interpret, enabling users to incorporate AI-assisted analysis into existing coaching workflows. Consequently, the framework satisfies the operational requirements of educational institutions, football academies, and amateur clubs.

4.8.3 Economic Feasibility

The proposed framework is economically feasible because it relies almost entirely on open-source software technologies. Python, OpenCV, PyTorch, YOLOv11, ByteTrack, Flask, PostgreSQL, and Visual Studio Code are freely available, eliminating software licensing costs.

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